The Primacy Bias In Deep Reinforcement Learning

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Primacy in Human Learning

- "Steve is impulsive, critical, and smart"
- "Steve is smart, critical, and impulsive"

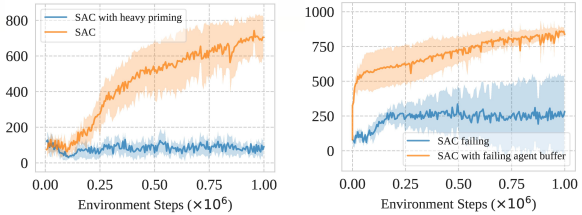
Same information, different perception

The Primacy Bias in Deep RL

A tendency to overfit early experiences that damages the rest of the learning process

Controlled Experiments

- Severe overfitting might be unrecoverable
- Data from an overfitted agent are sufficient



How can an agent properly learn from the data it collected?

Addressing the Primacy Bias

Periodically reset the agent's last layers while preserving its replay buffer

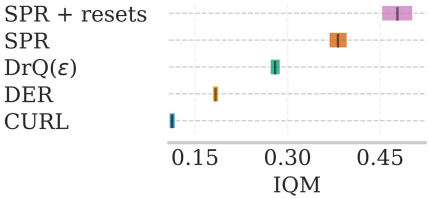
Contributions

- Analyze a special form of overfitting
- A simple remedy: resets
- Large performance gains
- Unlock novel data-efficient training regimes

Results

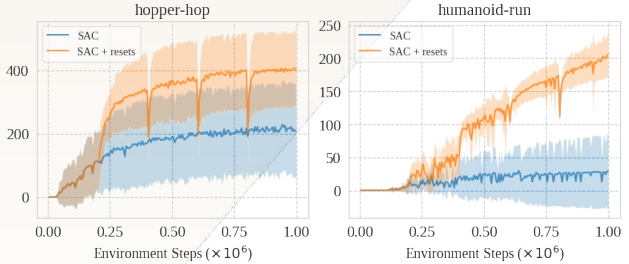
- Benchmarks
  - DMC-dense (SAC)
  - DMC-pixels (DrQ)
  - Atari 100k (SPR)
- Resets are a special form of regularization for the primacy bias

Method	IQM
SAC	501 (389, 609)
SAC + resets	**656** (549, 753)
SAC + dropout	219 (160, 285)
SAC + L2	412 (299, 524)
DrQ	569 (475, 662)
DrQ + resets	**762** (704, 815)
DrQ + dropout	492 (414, 567)
DrQ + L2	463 (362, 566)

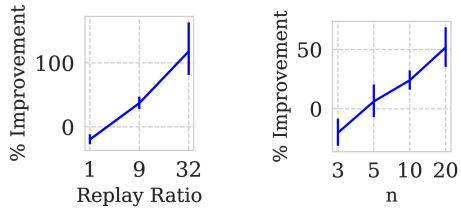


Analysis of Learning with Resets

The agent is able to recover incredibly quickly



Resets are more beneficial with high RR and *n*



Ablations

- Reset depth depends on representation complexity
- Resets fix TD collapses and divergence
- Resetting the optimizer does not matter
- Agents are robust to the number of resets